

Question number	Answer	Additional guidance	Mark
3(a)	A description that includes two of the following points: • (complex) transported to the Golgi (apparatus / body) (in vesicles) (1) • transported in vesicles (from the Golgi through the cytoplasm) to the (surface) membrane (1) • (released onto surface) by fusion of vesicles with membrane (1)	ACCEPT exocytosis from the (surface) {membrane / cell}	(2)

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3(b)(i)	0.37 (nm) (1)	3.7×10^{-1}	(1)

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3(b)(ii)	the microscope is able to distinguish two lines if they are {3.7 Å / 0.37 nm} or more apart (1)	ACCEPT value given in (i) structures / objects / points different / separate e.g minimum distance of 3.7 angstroms required for two things to be seen apart if two objects are closer together than 3.7 angstroms they appear as a single object IGNORE particles resolve / clear / more detail / quality / focus DO NOT ACCEPT magnification	(1)

Question number	Answer	Additional guidance	Mark
3(c)	An explanation that includes two of the following points: • because antigen presentation is needed to activate the T killer cells (1) • so that T killer cells will divide (1) • so that there are {enough / many} T killer cells to {destroy / lyse / release perforins onto} the host-infected cells (1)	ACCEPT cytotoxic T cells / CD8 cells throughout ACCEPT initiate CMI / clonal selection ACCEPT clone / clonal expansion ACCEPT chemicals / enzymes / proteases cause apoptosis	(2)

Question number	Answer	Additional guidance	Mark
3(d)	An explanation that includes the following points: • T killer cells {destroy / lyse / perforate} the {infected / host} cells (1) • releasing {viruses / viral components} (1) • so that macrophages can destroy them (1)	ACCEPT pathogen but not bacteria IGNORE damaged DO NOT ACCEPT virus / antigen ACCEPT phagocytes / phagocytosis of virus / virus engulfed / virus digested virus particles will not assemble (if components not inside a cell) DO NOT ACCEPT kills virus / T killer cells phagocytose / antibodies destroy	(3)

